UGOCHUKWU KINGSLEY NNAEMEKA

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PROFESSIONAL SUMMARY

Biochemist and M.Sc. researcher with experience in enzyme purification, biochemical characterization, molecular techniques, teaching, and Python-based systems biology modelling. Interested in integrating experimental biochemistry with computational approaches to biological research.

EDUCATION

M.Sc., Biochemistry and Molecular Biology Apr 2025 – Present
Obafemi Awolowo University, Ile-Ife, Nigeria

Research: Purification and biochemical characterization of rhodanese from a cyanide-tolerant bacterium isolated from cassava-processing soil.

B.Sc., Biochemistry - Second Class Upper Honours Oct 2018 – Jul 2023
Imo State University, Owerri, Nigeria | CGPA: 4.18/5.00

RESEARCH EXPERIENCE

Postgraduate Researcher, Enzymology Laboratory Apr 2025 – Present
Obafemi Awolowo University, Department of Biochemistry and Molecular Biology

- Investigate rhodanese production, purification, enzyme kinetics and physicochemical properties relevant to cyanide detoxification.
- Perform enzyme activity and protein assays; analyze purification, stability and biochemical characterization data.

Research Assistant Aug 2023 – Apr 2024
Department of Biochemistry, Imo State University

- Supported literature review, experimental design and phytochemical investigations involving onion skin extracts.

Student Researcher / Research Intern Feb – Oct 2022
Nigerian Institute of Medical Research, Lagos

- Gained practical exposure to DNA/RNA extraction, PCR, gel electrophoresis and laboratory sample preparation.

COMPUTATIONAL RESEARCH

Python-Based Systems Biology Project 2026
Neuronal Energy Metabolism and ATP Demand |
github.com/KingsleyUgochukwu/brain_energy_metabolism

- Developing Python-based models to explore neuronal energy metabolism, ATP demand and metabolic responses.

TEACHING EXPERIENCE

Biology Teacher (NYSC) Apr 2024 – Mar 2025

Oduduwa College, Ile-Ife, Nigeria

- Taught biology to approximately 108 students, supervised practicals and assessed learning outcomes.

Biology Tutor Aug 2023 – Mar 2024

Imo State University, Undergraduate Tutorial Centre

- Delivered introductory biology and biochemistry tutorials and supported over 100 students with exam preparation.

PUBLICATION

Ekeanyanwu, C. R., Ekeanyanwu, C. L., & Ugochukwu, K. N. (2024). Towards a natural treatment for mania: Red onion husk extract modulates neuronal resilience, redox signalling, and glial activation. International Journal of Bipolar Disorders, 12, Article 16. https://doi.org/10.1186/s40345-024-00338-7

PRESENTATION

Postgraduate College week October 2026

- Title: Biochemical Characterization of Purified Rhodanese from Mammaliicoccus sciuri isolated from a Cassava Processing Site
- Description: Poster presentation on production and purification of rhodanese enzyme from M. sciuri.

TECHNICAL SKILLS

- Laboratory: Enzyme purification and characterization; enzyme kinetics; UV–visible spectrophotometry; SDS-PAGE and native PAGE; chromatography; PCR and gel electrophoresis.
- Computational: Python, NumPy, pandas, SciPy, Matplotlib, data visualization and statistical analysis.

HONOURS AND LEADERSHIP

- Top-five graduating student, Biochemistry department 2023
- Faculty President, Faculty of Biological Sciences 2021–2022
- Departmental Academic Director 2020–2021